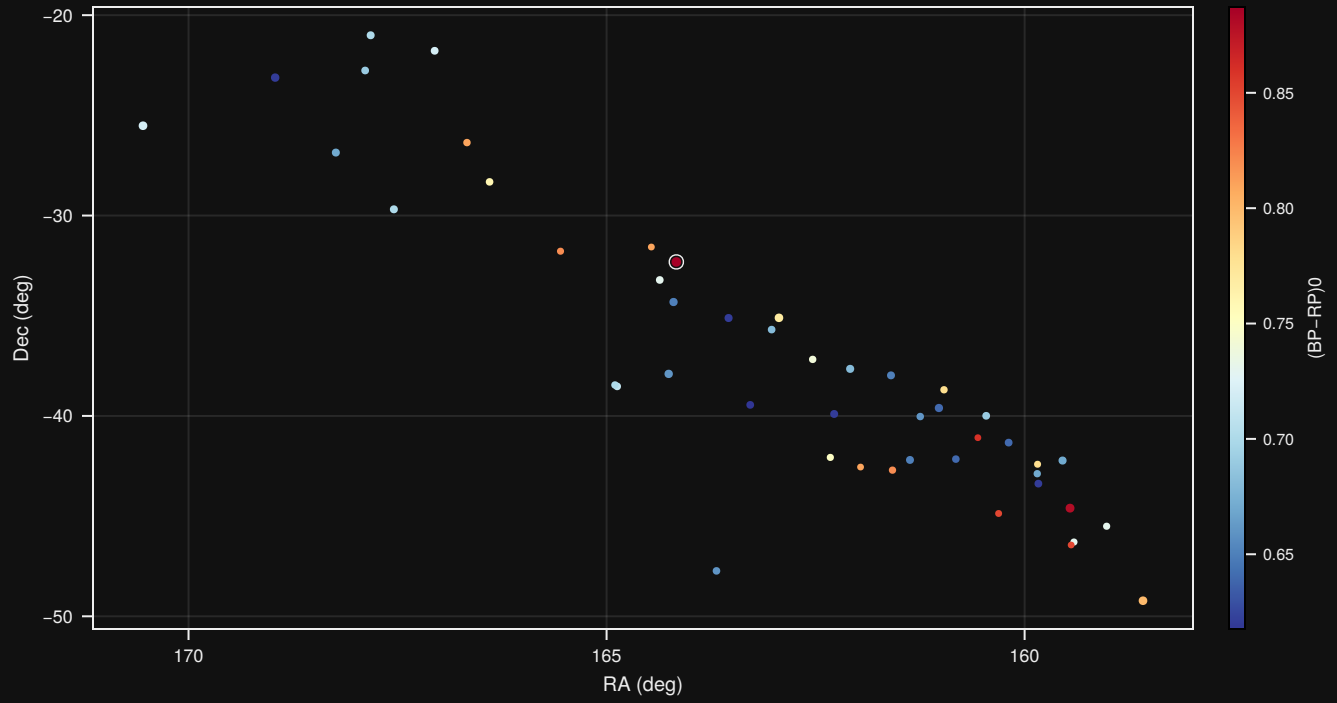
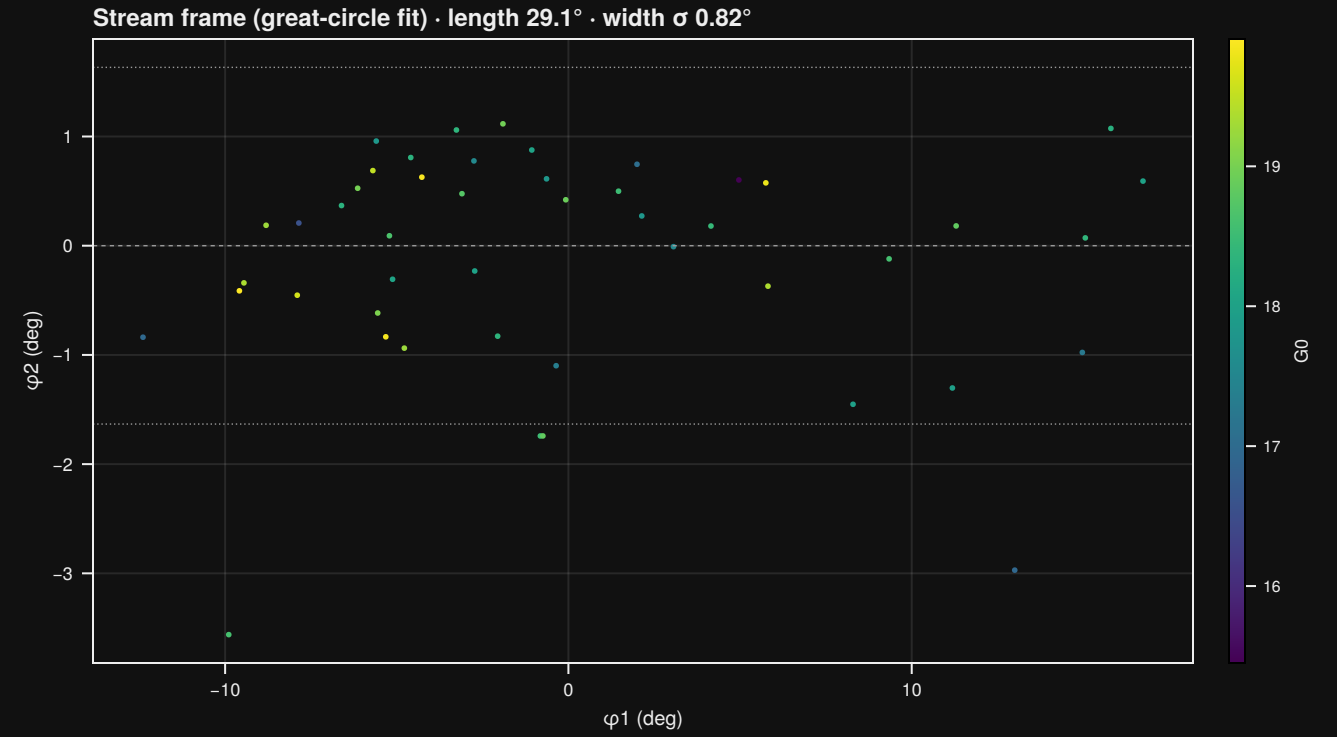
New-13

MEMBERSHIP	
Stars in catalog	47
with Gaia GSP-Phot	31 (66%)
with radial velocity	1
Stream id (sID)	38
GEOMETRY · great-circle fit	
Centroid RA, Dec	163.0, -38.7 deg
Centroid l, b	278.1, 17.9 deg
Pole RA, Dec	84.0, 14.0 deg
Length (φ1, 2–98%)	29.1 deg
Width σ(φ2), robust	0.82 deg
KINEMATICS	
pmRA* med ± σ	-5.62 ± 1.47 mas/yr
pmDE med ± σ	-9.49 ± 3.02 mas/yr
μφ1* med ± σ	-10.62 ± 3.06 mas/yr
μφ2 med ± σ	2.45 ± 0.99 mas/yr
Parallax (cat.) med	0.180 ± 0.218 mas
Parallax (wtd, ZP-corr.)	0.222 ± 0.022 mas
Distance	d = 4.5 ± 0.5 kpc
Physical length	2.29 kpc
RV median ± σ(MAD)	478.6 ± 0.0 km/s
RV gradient dV/dφ1	—
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.4–19.9 (18.4)
(BP−RP)0 median	0.70
GSP-Phot [M/H] (n=31)	-1.88 ± 0.53
GSP-Phot Teff median	5753 K
RUWE > 1.4	3.2 %
PM clusters (largest)	4 (12 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	47 | 47
n_RV catalog | paper	1 | 1
<[Fe/H]> paper	-1.19
Stellar mass (paper)	250 Msun
Pole offset vs paper	0.1 deg
Remark	High velocity

Sky map · catalog members (n = 47) · size ∝ brightness · ring = has RV



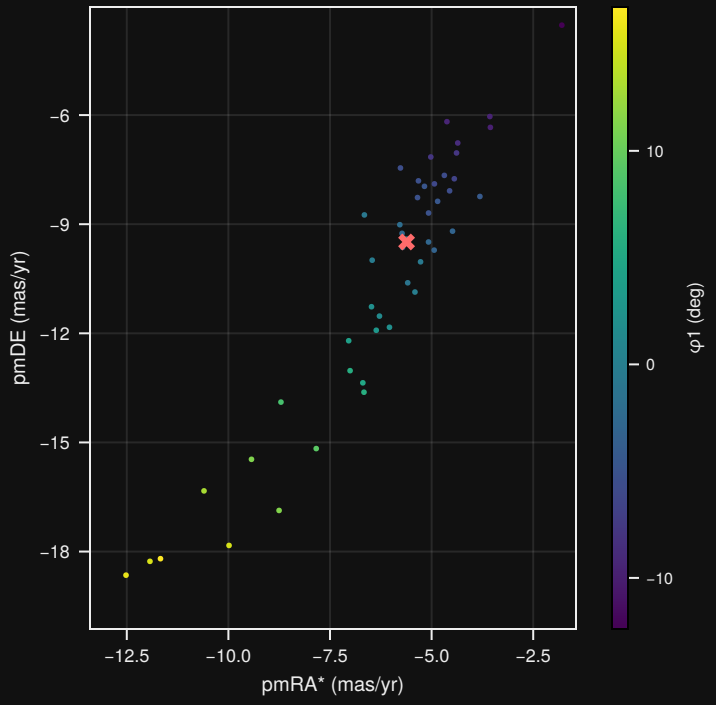
Stream frame (great-circle fit) · length 29.1° · width σ 0.82°



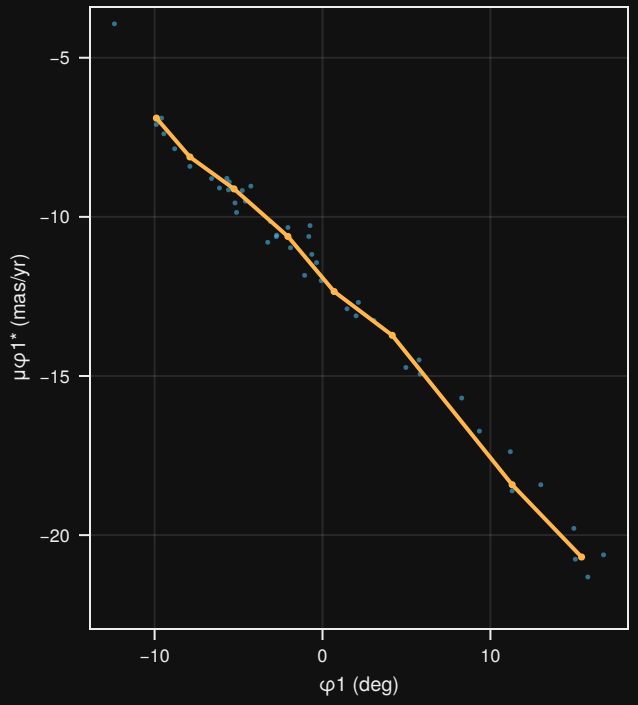
New-13

Kinematics · proper motion · parallax · radial velocity

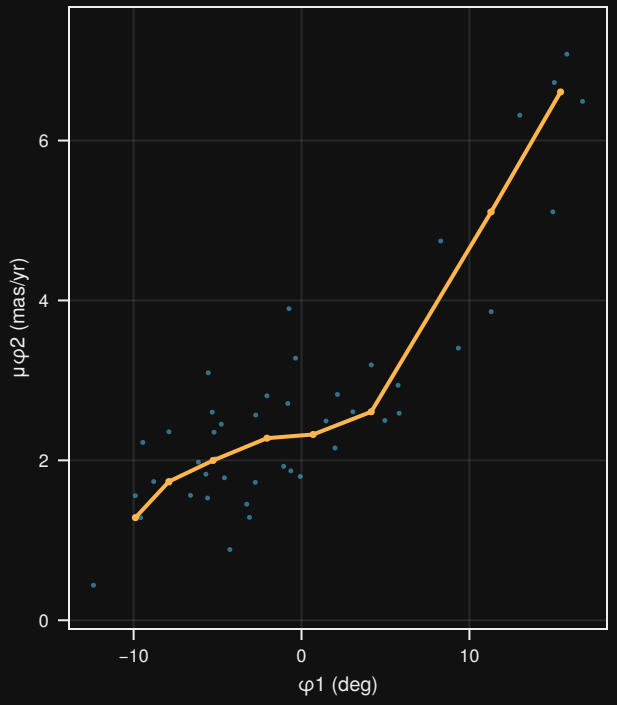
Proper motion (n = 47)



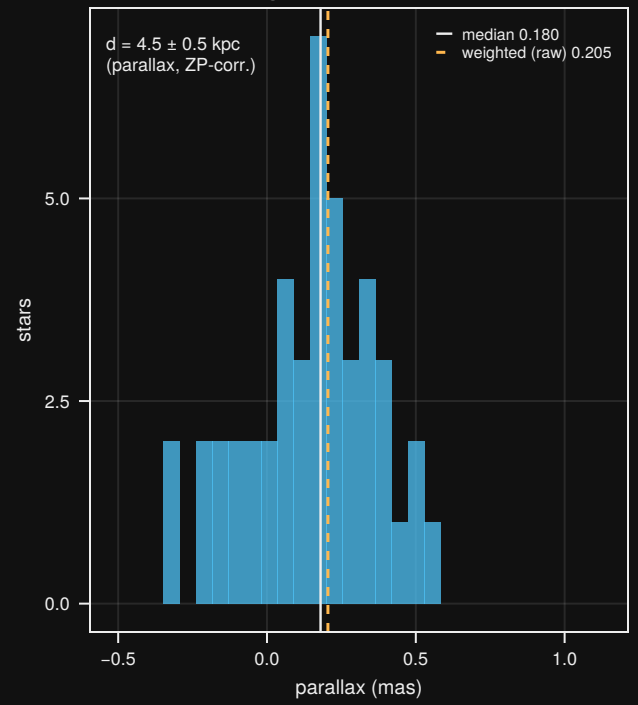
μφ1* vs φ1 · σ = 3.06



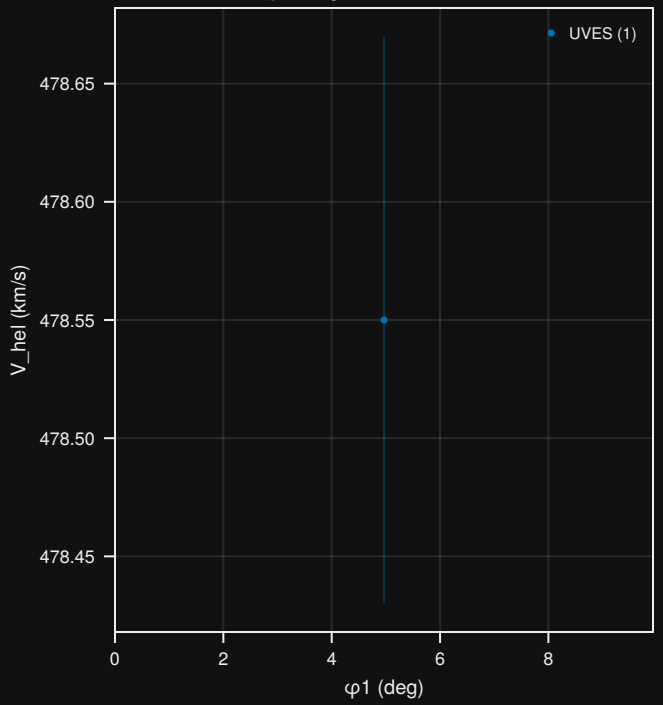
μφ2 vs φ1 · σ = 0.99



Parallax (catalog, n = 47)



Radial velocity vs φ1 (n = 1)

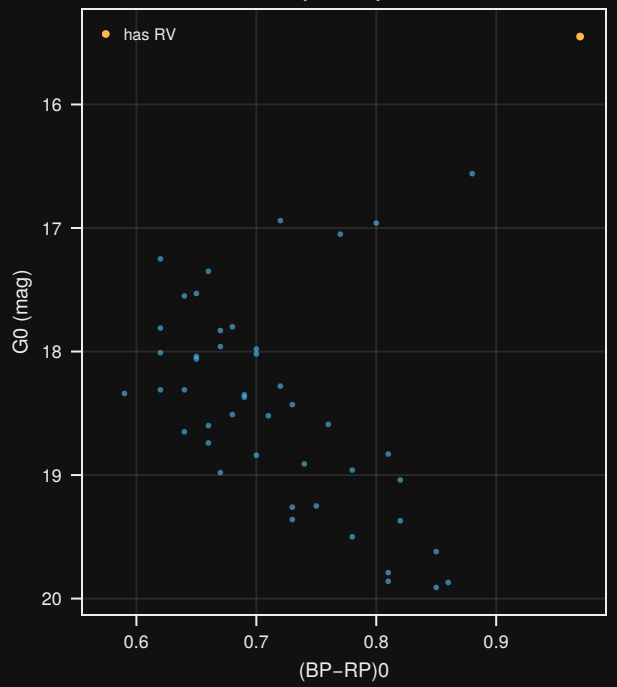


V_hel distribution

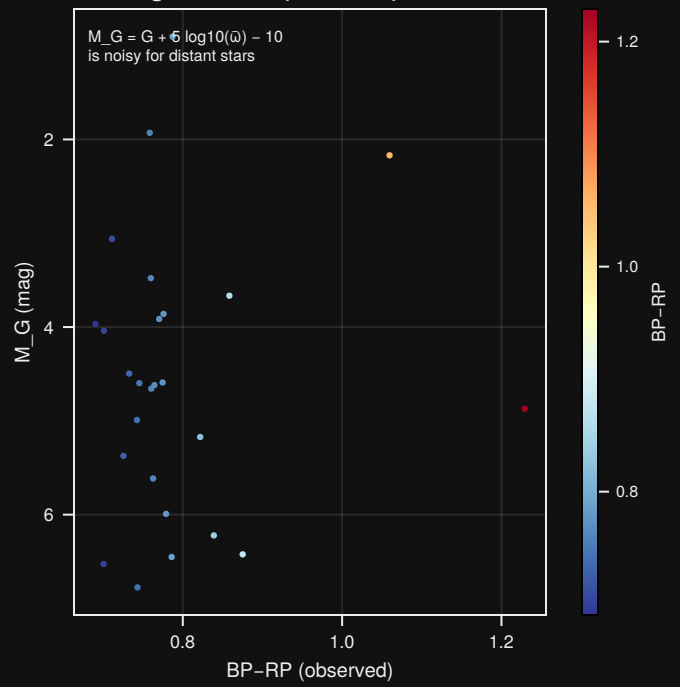
n_RV = 1 (< 3)

New-13

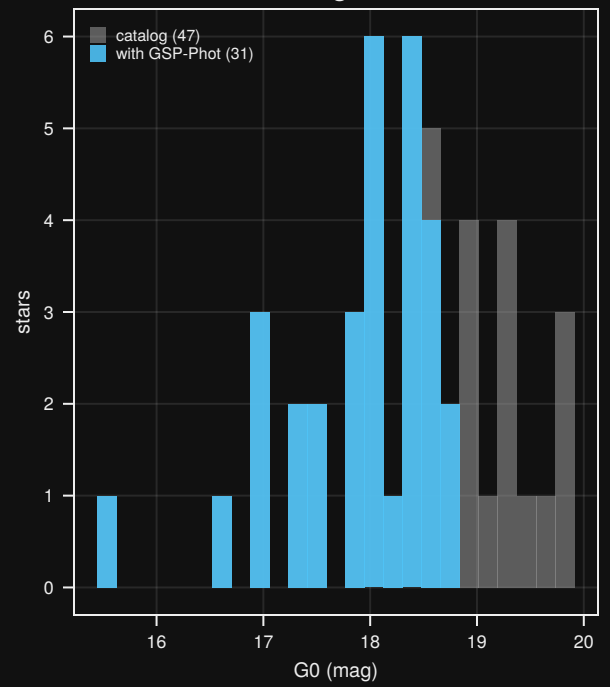
CMD · dereddened (n = 47)



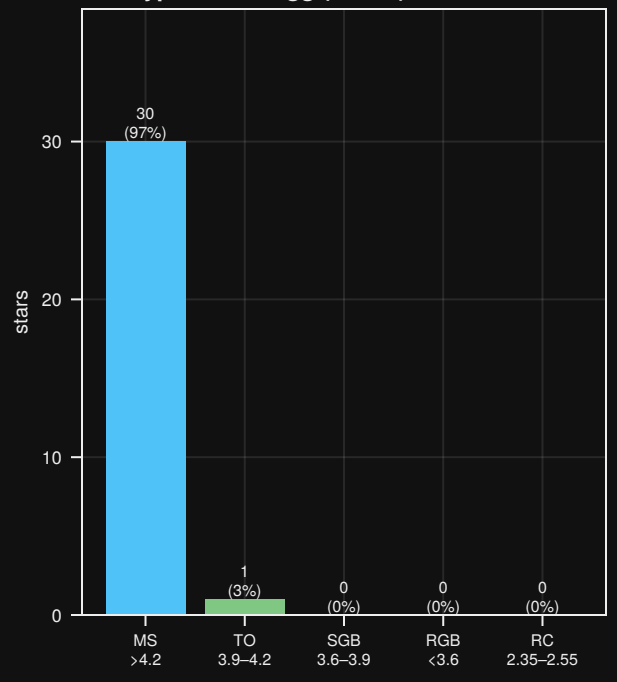
HR diagram · ϖ>0 (n = 26/31)



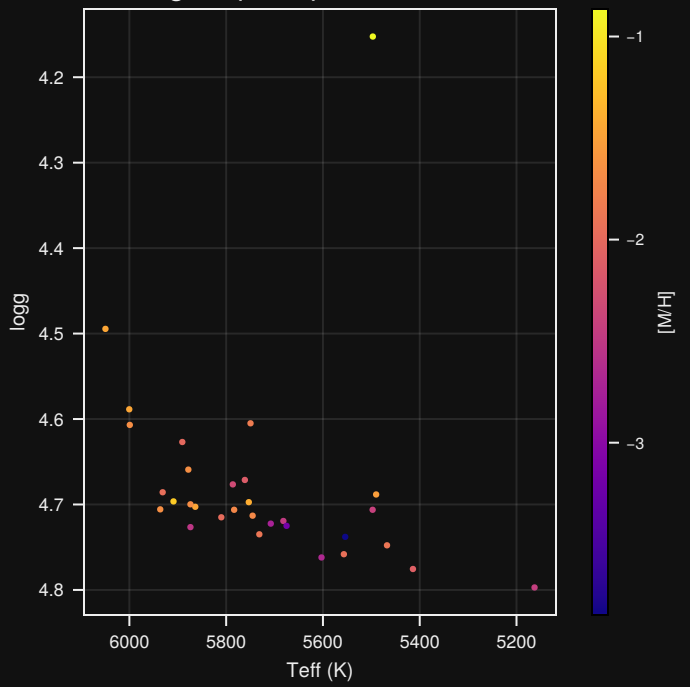
G0 & GSP-Phot coverage



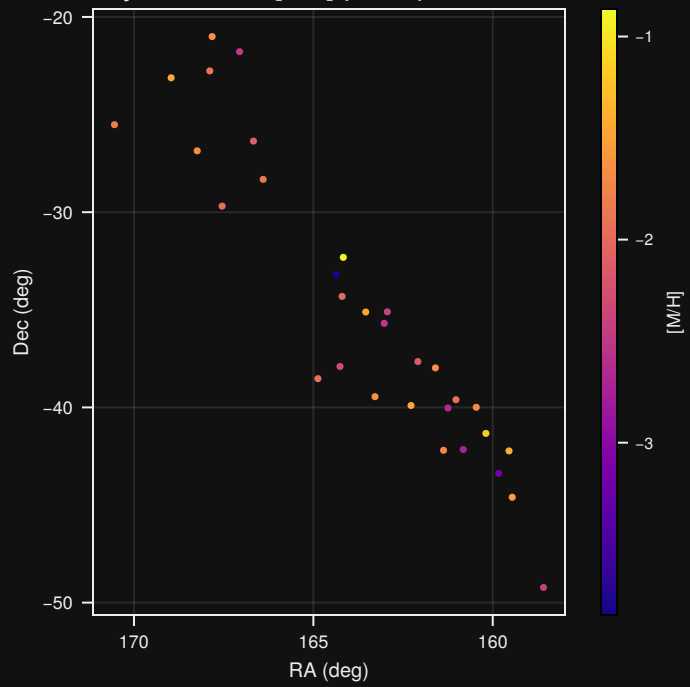
Star types from logg (n = 31)



Kiel diagram (n = 31)



Sky · GSP-Phot [M/H] (n = 31)

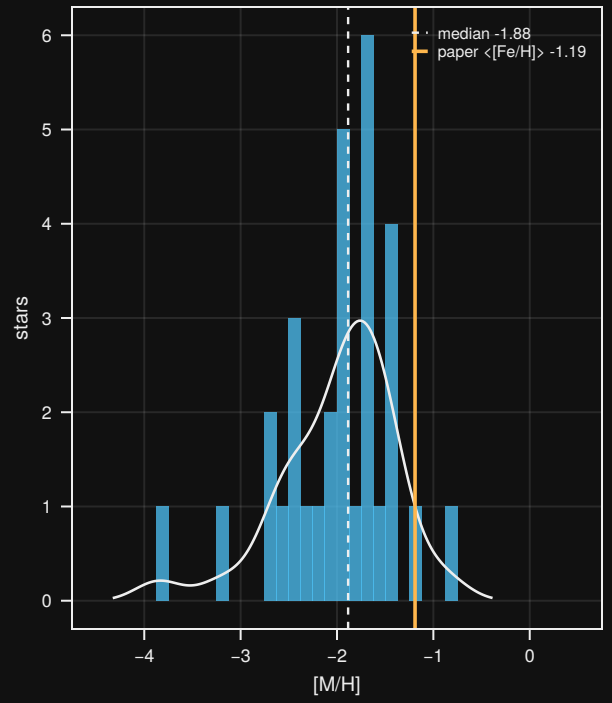


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

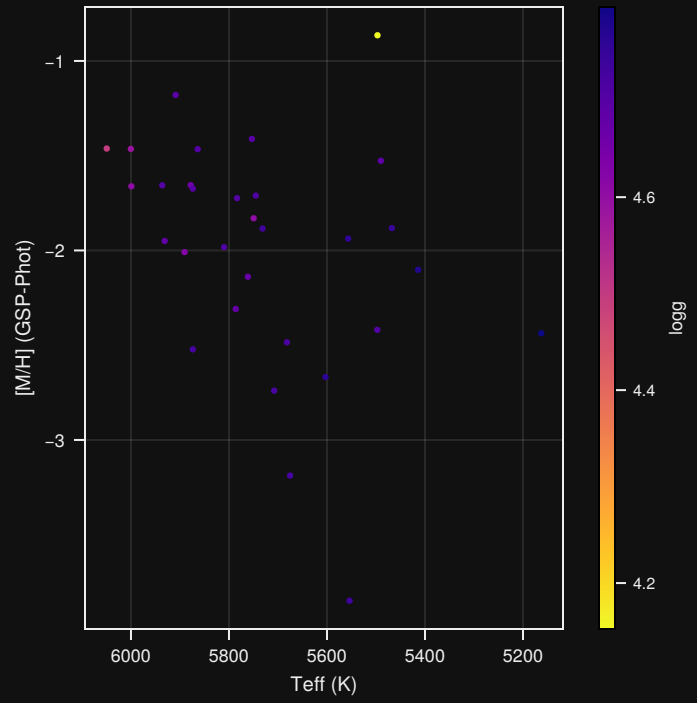
New-13

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 31)



Teff vs [M/H] (n = 31)



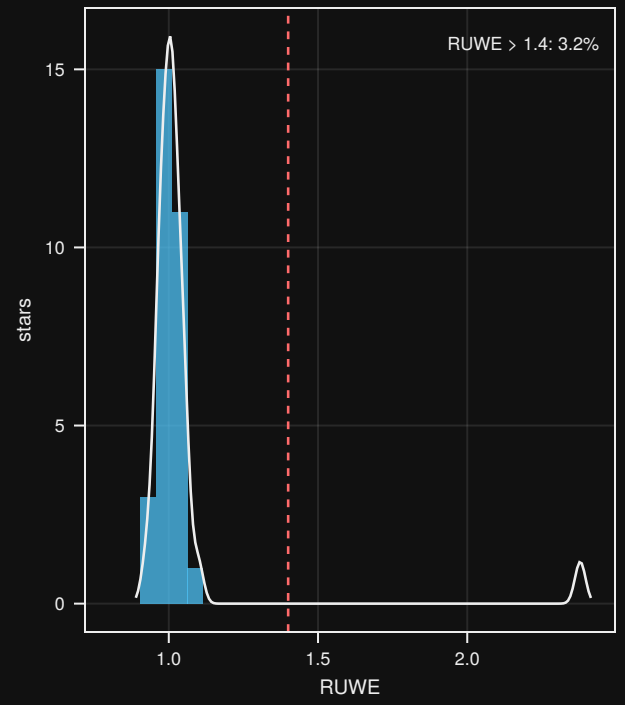
FLAME age (n = 0)

FLAME ages for only 0 star(s)
(FLAME needs good parallaxes)

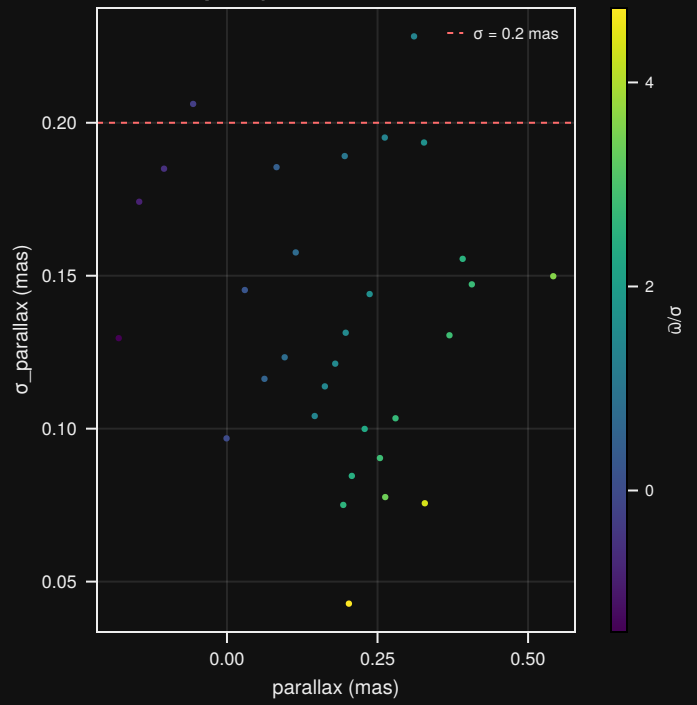
Age–metallicity (n = 0)

n < 3

RUWE (n = 31)

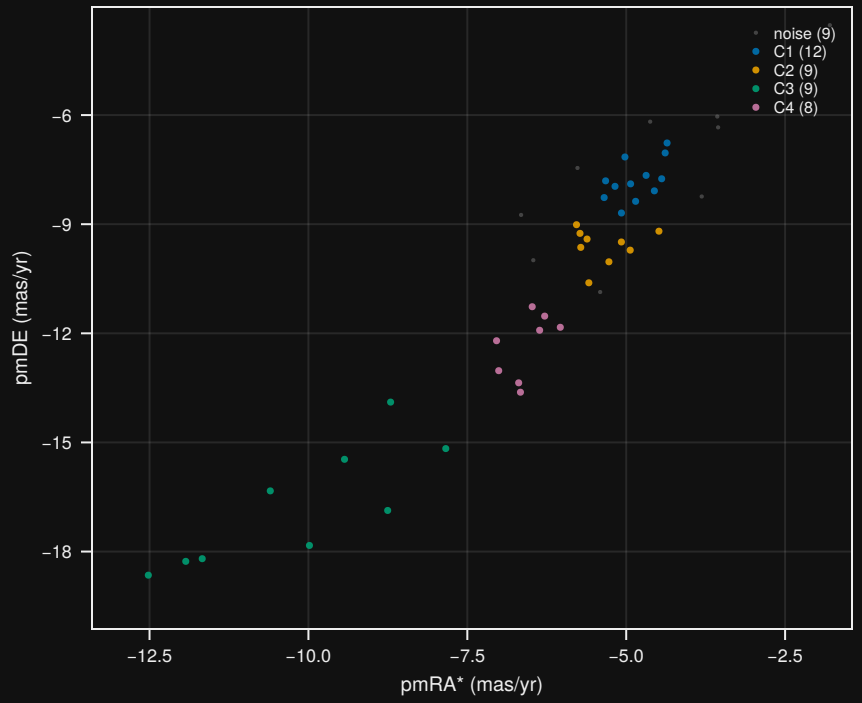


Parallax quality (n = 31)

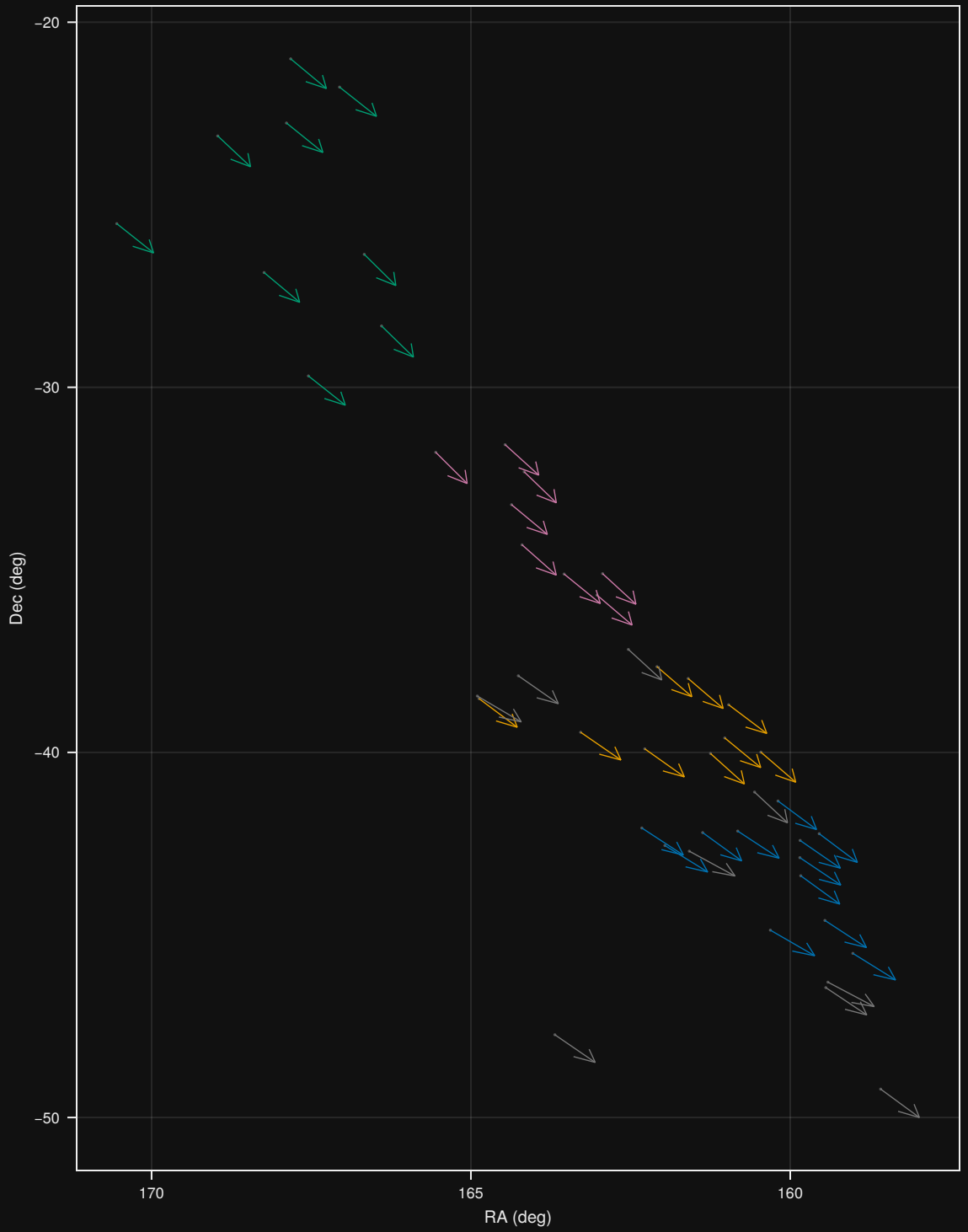


New-13

PM clustering · HDBSCAN · 4 cluster(s) · noise 19%



Proper-motion directions (arrows) coloured by cluster



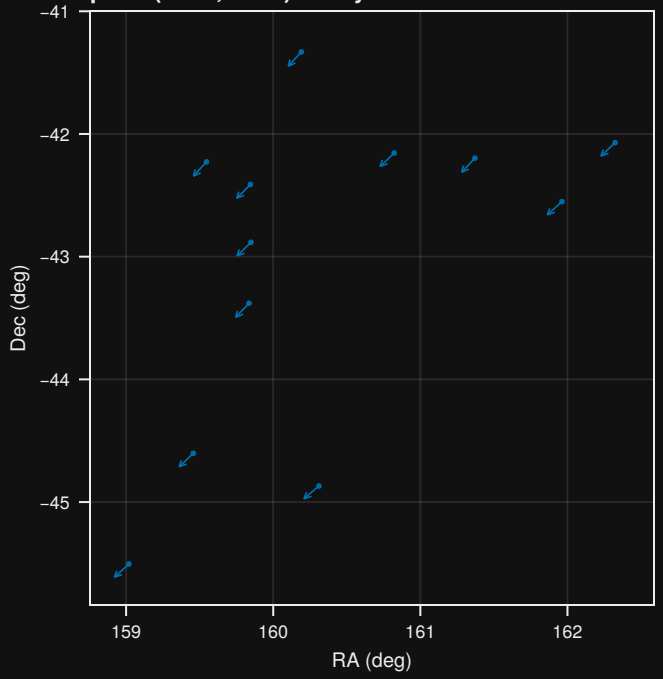
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	9	-4.62	-7.45	1.58	-2.37	0
C1	12	-4.89	-7.85	0.46	-1.63	0
C2	9	-5.59	-9.49	0.29	-1.82	0
C3	9	-9.98	-16.87	1.89	-1.88	0
C4	8	-6.57	-12.06	0.37	-2.21	1

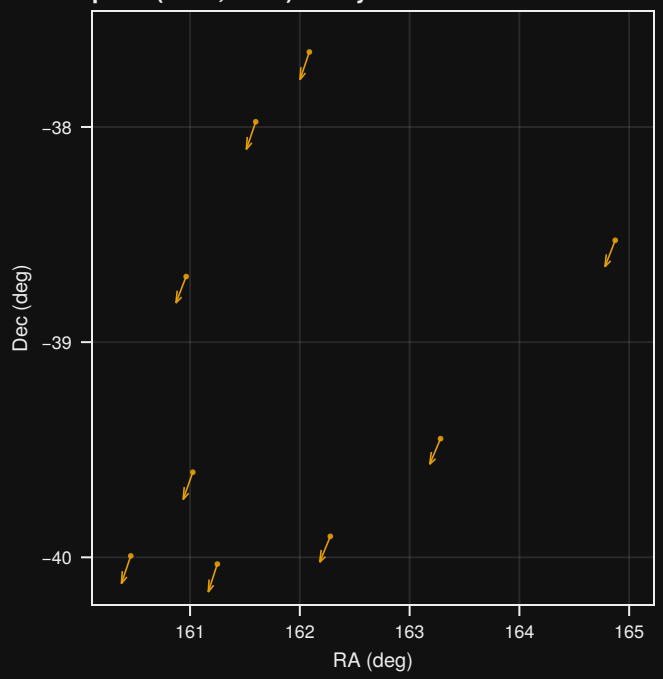
New-13

Per-cluster proper-motion vector maps

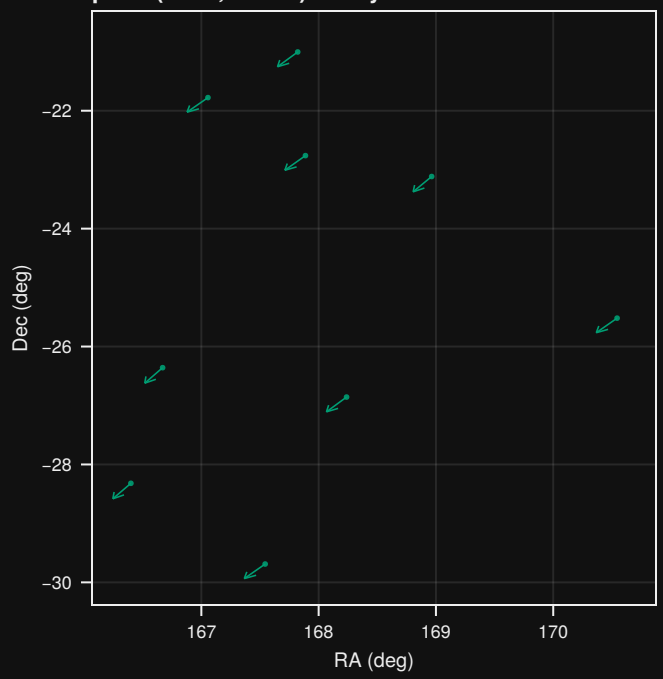
Cluster C1 · n = 12
pm = (-4.89, -7.85) mas/yr



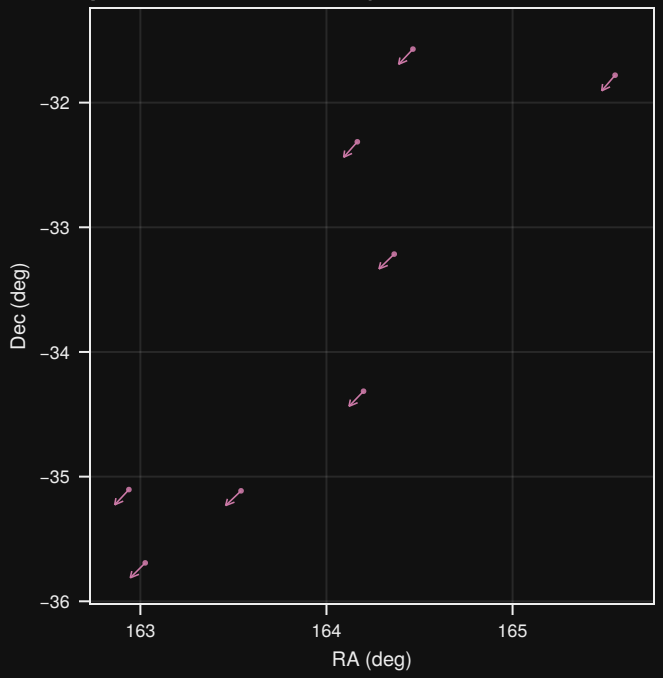
Cluster C2 · n = 9
pm = (-5.59, -9.49) mas/yr



Cluster C3 · n = 9
pm = (-9.98, -16.87) mas/yr



Cluster C4 · n = 8
pm = (-6.57, -12.06) mas/yr



Noise · n = 9

